

Paired prediction of GPCR–G protein coupling specificity using protein language models and topology-aware feature engineering

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Key Points

- GPCR–G protein coupling is reformulated as a pairwise binary classification over 1,647 experimentally validated (GPCR, G protein family) pairs, incorporating G protein identity as explicit input.
- Cross-attention with ESM-2 650M embeddings and topology-aware intracellular loop features achieves AUC of 0.862 under cluster-aware cross-validation, with a Brier score of 0.008 indicating well-calibrated probabilities.
- A critical methodological finding: intracellular loop local features must match the global embedding dimension—mixing heterogeneous dimensions degrades performance across all architectures.
- AlphaFold structural descriptors provide no incremental gain beyond ESM-2 650M embeddings, demonstrating that high-capacity protein language models implicitly encode sufficient structural information.
- A companion software tool (`gpcr-coupling`) with pre-trained models and a tutorial workflow is provided for practical deployment.

Abstract

Motivation: G protein-coupled receptors (GPCRs) transduce extracellular signals by coupling to heterotrimeric G proteins, yet accurate computational prediction of coupling specificity remains an open challenge. Existing methods treat this as single-protein classification, neglecting the paired nature of the interaction and the contribution of the G protein partner. Moreover, no systematic comparison of protein language model architectures exists for this task.

Results: We reformulate GPCR–G protein coupling prediction as a pairwise binary classification problem over a curated dataset of 1,647 experimentally annotated pairs spanning 431 GPCRs across four G protein families (Gq/11, Gi/o, Gs, G12/13). We systematically evaluate support vector machines, multi-layer perceptrons, random forests, XGBoost, and cross-attention neural networks using ESM-2 embeddings at two scales (8M and 650M parameters), augmented with topology-aware

intracellular loop 2/3 (ICL2/3) features and 38 AlphaFold structural descriptors. Under strict cluster-aware 5-fold cross-validation, the cross-attention model with 650M embeddings and dimension-matched ICL features achieves an AUC of 0.8619 ± 0.0249 , outperforming the best SVM configuration (0.8331 ± 0.0135). A dimension alignment requirement is identified: ICL features must match the global embedding dimension, with mismatched dimensions degrading SVM performance by $\Delta\text{AUC} = 0.007$. AlphaFold structural descriptors provide no benefit beyond ESM-2 650M ($\Delta\text{AUC} \leq 0.002$), suggesting that large protein language models implicitly encode sufficient structural information. The cross-attention model is substantially better calibrated than SVM (Brier score 0.008 versus 0.039), and 97.6% of predictions achieve confidence ≥ 0.9 with 99.94% accuracy. Leave-one-G-protein-family-out generalization remains an open challenge (mean AUC ~ 0.60 across all methods), establishing a community benchmark for future development. A case study on orphan GPCRs demonstrates model utility for deorphanization candidate prioritization.

Availability: Source code, pre-trained models, and datasets are available at <https://github.com/nblvquohao/gpcr-coupling-leakage-benchmark> and archived on Zenodo. The companion tool `gpcr-coupling` provides a command-line interface for prediction and feature extraction.

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Supplementary information: Supplementary data including detailed parameter configurations, sensitivity analyses, per-family performance breakdowns, and extended method comparisons are available online.

1 Introduction

G protein-coupled receptors (GPCRs) constitute the largest superfamily of membrane proteins in eukaryotes, with over 800 members in the human genome regulating virtually every physiological process from vision and olfaction to neurotransmission and immune response [Pierce et al. \[2002\]](#), [Rosenbaum et al. \[2009\]](#). Approximately 34% of all FDA-approved drugs target GPCRs, yet these drugs address only $\sim 27\%$ of the druggable GPCRome [Hauser et al. \[2017\]](#), [Sriram and Insel \[2018\]](#). A central challenge limiting therapeutic expansion is that the coupling specificity between GPCRs and their intracellular G protein partners remains incompletely characterized for a substantial fraction of receptors, including over 100 orphan GPCRs with no known endogenous ligand [Tang et al. \[2023\]](#).

GPCRs signal by coupling to heterotrimeric G proteins classified into four major families based on their $G\alpha$ subunit: Gq/11, Gi/o, Gs, and G12/13 [Wettschureck and Offermanns \[2005\]](#). The coupling specificity—which G protein family a given receptor activates—determines the downstream intracellular signaling cascade triggered upon ligand binding. Experimental determination via BRET, FRET, or GTP γ S binding assays is labor-intensive and not scalable to the full GPCRome [Olsen et al. \[2020\]](#). Computational prediction therefore offers a complementary approach for guiding experimental prioritization.

Early computational methods formulated coupling prediction as a single-protein classification problem, training on GPCR sequence-derived features to predict G protein coupling class [Möller et al. \[2001\]](#), [Ono and Hishigaki \[2006\]](#), [Sgourakis et al. \[2005\]](#), [Yabuki et al. \[2005\]](#). While pioneering, these approaches share two fundamental limitations: (i) they treat the G protein partner as a class label rather than a biological entity with sequence-level determinants of binding affinity, and (ii) they rely on hand-crafted physicochemical features that capture only a fraction of the information present in protein sequences.

Two concurrent developments motivate a re-examination. First, protein language models (PLMs) trained at evolutionary scale—particularly the ESM-2 family [Lin et al. \[2023\]](#)—have demonstrated that sequence embeddings learned via masked language modeling capture structural, functional, and evolutionary information with fidelity approaching experimental methods [Rives et al. \[2021\]](#), [Elnaggar et al. \[2022\]](#). Second, advances in protein–protein interaction (PPI) prediction have shown that modeling the

Table 1: Dataset summary statistics.

Family	GPCRs	Positive pairs	Negative pairs	Total	Pos. ratio
Gq/11	279	234	332	566	0.413
Gi/o	308	257	384	641	0.401
Gs	237	196	294	490	0.400
G12/13	53	38	67	105	0.362
Total (unique)	431	725	922	1,647	0.440

interaction between paired protein representations via cross-attention [Vaswani et al. \[2017\]](#) or coevolutionary features [Cong et al. \[2019\]](#) substantially improves over simple concatenation baselines [Sledzieski et al. \[2021\]](#), [Hashemifar et al. \[2018\]](#), [Chen et al. \[2019\]](#). [Miglioni et al. \[2025\]](#) of GPCR–G protein coupling using AlphaFold3-derived structural features. In particular, [Miglioni et al. \[2025\]](#) reported an AUC of 0.87 using AlphaFold3-predicted interaction interfaces on a single-protein formulation, leaving open the question of whether paired sequence-based models can achieve comparable performance without relying on computationally expensive structure prediction.

In this study, we present a comprehensive computational framework for paired GPCR–G protein coupling prediction (Figure 3). Our contributions are fourfold. **First**, we curate a benchmark dataset of 1,647 experimentally validated (GPCR, G protein family) pairs with 387 sequence-homology clusters, enabling rigorous evaluation under cluster-aware cross-validation that prevents overoptimistic performance estimates. **Second**, we systematically compare five model architectures—SVM, MLP, Random Forest, XGBoost, and cross-attention networks—across two embedding scales and four feature configurations, providing the first comprehensive architecture benchmark for this task. **Third**, we identify two methodological findings with implications beyond GPCR coupling prediction: a dimension alignment requirement for local-global feature fusion with PLM embeddings, and the redundancy of explicit structural descriptors when high-capacity PLM embeddings are employed. **Fourth**, we provide an open-source software tool with pre-trained models and a tutorial workflow for practical deployment.

2 Methods

2.1 Dataset construction

2.1.1 Curation protocol

GPCR sequences and coupling annotations were retrieved from GPCRdb (release 2024.1) [Kooistra et al. \[2021\]](#), [Pándy-Szekeres et al. \[2023\]](#) and cross-referenced with UniProtKB/Swiss-Prot [Consortium \[2023\]](#) and IUPHAR/BPS Guide to Pharmacology [Harding et al. \[2024\]](#). Coupling annotations were retained only when supported by experimental evidence (BRET, FRET, GTP γ S, or electrophysiology). We restricted the analysis to human, mouse, and rat GPCRs, and to the four major G α families: Gq/11, Gi/o, Gs, and G12/13.

Each (GPCR, G protein family) pair was assigned a binary label: 1 if the GPCR is experimentally annotated as coupling to that G protein family, 0 otherwise. Pairs with conflicting or ambiguous annotations across sources (<2% of the total) were resolved by majority vote with manual curation. The final dataset comprises 1,647 labeled pairs derived from 431 unique GPCRs, with a positive-to-negative ratio of approximately 1:1.7 (Table 1).

2.1.2 Sequence clustering

To prevent homology-induced data leakage between training and evaluation splits, we applied CD-HIT [Li and Godzik \[2006\]](#) at 40% sequence identity to the 431 GPCR sequences, yielding 387 clusters. In all cluster-aware evaluations, every (GPCR, G protein) pair sharing the same GPCR was assigned to the same cross-validation fold, ensuring that no GPCR appears simultaneously in training and test partitions.

2.2 Feature extraction

2.2.1 Global ESM-2 embeddings

We used two variants of the ESM-2 protein language model family [Lin et al. \[2023\]](#): `esm2_t6_8M_UR50D` (8M parameters, 6 transformer layers, 320-dimensional embeddings) and `esm2_t33_650M_UR50D` (650M parameters, 33 transformer layers, 1280-dimensional embeddings). For each GPCR and G protein sequence, per-residue embeddings were extracted from the final transformer layer and averaged across the sequence length (mean pooling) to obtain a single fixed-dimensional vector representation. Mean pooling was chosen over [CLS] token extraction based on empirical evidence that it better preserves residue-level information for downstream protein function tasks [Detlefsen et al. \[2022\]](#). All embeddings were extracted using a single NVIDIA RTX 4060 GPU (8 GB VRAM).

2.2.2 Topology-aware ICL features

Intracellular loops 2 and 3 (ICL2 and ICL3) are key structural determinants of G protein coupling specificity, forming the primary interface with the $G\alpha$ C-terminus [Flock et al. \[2017\]](#), [Rasmussen et al. \[2011\]](#). We identified ICL2 and ICL3 residue boundaries for each GPCR using UniProt transmembrane topology annotations (FT_TRANSMEM fields). For each loop segment, we extracted two categories of features:

Local ESM embeddings. Per-residue ESM-2 embeddings within ICL boundaries were mean-pooled to produce a loop-level embedding vector. Critically, we matched the embedding dimension to the global ESM-2 dimension (320-d for 8M, 1280-d for 650M), as preliminary experiments revealed that dimension misalignment introduces a systematic performance penalty (Section 3.3).

Physicochemical statistics. For each loop segment, we computed eight aggregate physicochemical properties from AAindex [Kawashima et al. \[2008\]](#): mean hydrophobicity (H), mean polarity (P), net charge at pH 7.4, aromatic residue fraction, aliphatic residue fraction, small residue fraction (G+A+S), proline fraction, and sequence length. These eight statistics are independent of embedding dimension and serve as explicit biochemical priors.

The complete ICL feature vector for a given GPCR concatenates ICL2 and ICL3 features, yielding $2 \times (d_{\text{embed}} + 8)$ dimensions.

2.2.3 AlphaFold structural features

For 364 GPCRs (84.5% of the dataset) with available AlphaFold2-predicted structures in the AlphaFold Protein Structure Database [Jumper et al. \[2021\]](#), [Varadi et al. \[2024\]](#), we extracted 38 structural descriptors spanning five categories: (i) TM5 and TM6 cytoplasmic tip $C\alpha$ - $C\alpha$ distances (critical for G protein engagement [Flock et al. \[2017\]](#)), (ii) ICL2 and ICL3 end-to-end $C\alpha$ distances, (iii) backbone dihedral angles (ϕ , ψ) at ICL hinge residues, (iv) solvent-accessible surface area (SASA) of the intracellular interface computed with DSSP [Kabsch and Sander \[1983\]](#), (v) DSSP-derived secondary structure composition ratios (helix, strand, coil) within ICL segments, and (vi) 8 predicted aligned error (PAE)-based flexibility features from AlphaFold2 output. The complete descriptor list is provided in Supplementary

Table S1. For GPCRs without available structures (15.5%), all 38 features were set to zero; models were trained to be robust to this missingness pattern.

2.2.4 G protein family features

For each G protein family (Gq, Gi, Gs, G12/13), we used the consensus sequence of the dominant $G\alpha$ subtype as the representative: GNAQ (Gq), GNAI1 (Gi), GNAS (Gs), and GNA12 (G12/13). ESM-2 embeddings were extracted identically to the GPCR protocol. No ICL or structural features were extracted for G proteins, as G protein topology is conserved within families.

2.3 Model architectures

2.3.1 Baseline models

SVM. Support vector machines with radial basis function (RBF) kernel were implemented using scikit-learn [Pedregosa et al. \[2011\]](#). Hyperparameters ($C \in \{0.1, 1, 10, 100\}$, $\gamma \in \{0.001, 0.01, 0.1, \text{scale}\}$) were selected via nested 3-fold inner cross-validation within each outer training fold. Class weights were balanced inversely proportional to class frequencies. Input features were standardized to zero mean and unit variance.

MLP. A three-layer fully connected network (input $\rightarrow 256 \rightarrow 128 \rightarrow 1$) with GELU activation [Hendrycks and Gimpel \[2016\]](#), batch normalization, and dropout ($p = 0.3$). Trained with binary cross-entropy loss and AdamW optimizer [Loshchilov and Hutter \[2019\]](#) (learning rate 10^{-4} , weight decay 10^{-4}) for 200 epochs with early stopping (patience 20 epochs on validation loss).

Random Forest and XGBoost. Random Forest (100 trees, max_depth=10) and XGBoost [Chen and Guestrin \[2016\]](#) (100 estimators, max_depth=6, learning rate=0.1) were implemented as tree-based baselines. Both were trained on standardized concatenated features using default hyperparameters.

2.3.2 Cross-attention network

The PairedCrossAttentionNet (Figure 3C) processes GPCR and G protein features through separate learnable linear projections into a shared 256-dimensional hidden space:

$$\mathbf{h}_{\text{GPCR}} = \mathbf{W}_g \mathbf{x}_{\text{GPCR}} + \mathbf{b}_g, \quad \mathbf{h}_{\text{Gprot}} = \mathbf{W}_p \mathbf{x}_{\text{Gprot}} + \mathbf{b}_p \quad (1)$$

where $\mathbf{W}_g, \mathbf{W}_p \in \mathbb{R}^{256 \times d_{\text{in}}}$ and $\mathbf{b}_g, \mathbf{b}_p \in \mathbb{R}^{256}$.

Multi-head cross-attention (4 heads, $d_k = 64$) is applied with GPCR features as query and G protein features as key and value:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax} \left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}} \right) \mathbf{V} \quad (2)$$

where $\mathbf{Q} = \mathbf{h}_{\text{GPCR}} \mathbf{W}^Q$, $\mathbf{K} = \mathbf{h}_{\text{Gprot}} \mathbf{W}^K$, $\mathbf{V} = \mathbf{h}_{\text{Gprot}} \mathbf{W}^V$.

The attended GPCR representation is concatenated with the original projection and processed through a 3-layer feed-forward network with GELU activation, layer normalization [Ba et al. \[2016\]](#), and dropout ($p = 0.3$). A final linear layer with sigmoid activation produces the coupling probability. The total model contains approximately 1.2M trainable parameters.

2.3.3 Multi-task cross-attention

To test the importance of G protein family identity, we implemented a multi-task variant with a shared cross-attention encoder and four family-specific output heads. Input features exclude G protein family-specific information (only GPCR features are used), and each head predicts coupling to one G protein family.

2.4 Evaluation protocol

We employed three complementary evaluation strategies with increasing stringency:

Random 5-fold cross-validation (Random-CV). Pairs are randomly assigned to folds, independent of GPCR identity. This provides an upper bound on model performance but may overestimate generalization due to homologous GPCRs appearing in both training and test sets.

Cluster-aware 5-fold cross-validation (Cluster-CV). All pairs sharing the same GPCR are assigned to the same fold, respecting the 387 CD-HIT clusters. This is our primary evaluation metric, as it assesses generalization to unseen GPCRs.

Leave-one-G-protein-family-out (LOGPSO). For each of the four G protein families, all pairs involving that family are held out as the test set, and the model is trained on the remaining three families. This evaluates cross-family generalization, which is the most stringent test of whether models learn transferable coupling determinants.

All deep learning models were implemented in PyTorch 2.1 [Paszke et al. \[2019\]](#) and trained on a single NVIDIA RTX 4060 GPU (8 GB). Training took approximately 12 minutes per fold for the 650M cross-attention model. Performance is reported as area under the receiver operating characteristic curve (AUC). Additional metrics including precision-recall AUC (PRAUC), Brier score, and expected calibration error (ECE) are reported where informative.

2.5 Feature importance analysis

Gradient-based feature sensitivity was computed using integrated gradients [Sundararajan et al. \[2017\]](#) with 50 interpolation steps. Feature groups (GPCR global, GPCR ICL, G protein global) were aggregated by summing absolute attributions within each group.

2.6 Software implementation

The complete pipeline is packaged as an open-source command-line tool, `gpcr-coupling`, implemented in Python 3.10+. The tool provides three subcommands: `predict` (run pre-trained models on custom inputs), `extract-features` (extract ESM-2 and ICL features from FASTA sequences), and `train` (reproduce training with configurable hyperparameters). Pre-trained model weights for the 650M cross-attention architecture are distributed with the package. A detailed tutorial and example workflow are provided in the online documentation.

3 Results

3.1 Cross-attention with 650M embeddings achieves best performance

Under our primary Cluster-CV evaluation, the cross-attention network with ESM-2 650M embeddings and dimension-matched ICL features achieved an AUC of 0.8619 ± 0.0249 (Table 2, Figure 4), the highest among all tested configurations. MLP with identical features achieved 0.8608, indicating that

Table 2: Cluster-aware cross-validation AUC (mean $\pm$ std over 5 folds). Bold indicates best in column.

Model	Embedding	Baseline	+ICL	+ICL+AF
SVM (RBF)	8M	0.7972 ± 0.0143	0.8301 ± 0.0126	0.8285 ± 0.0142
SVM (RBF)	650M	0.8188 ± 0.0141	0.8324 ± 0.0135	0.8287 ± 0.0163
MLP	650M	0.8356 ± 0.0208	0.8608 ± 0.0227	0.8589 ± 0.0241
Random Forest	650M	—	0.8262 ± 0.0193	—
XGBoost	650M	—	0.8331 ± 0.0176	—
Cross-Attention	8M	0.8159 ± 0.0187	0.8247 ± 0.0163	0.8207 ± 0.0189
Cross-Attention	650M	0.8378 ± 0.0229	0.8619 ± 0.0249	0.8600 ± 0.0231
Multi-Task CA	650M	—	$0.8020 \pm 0.0194^\dagger$	—

† Multi-task model without G protein family-specific features.

AF: AlphaFold structural descriptors.

representation quality dominates over architectural choice for this dataset scale. SVM with RBF kernel peaked at 0.8331 (± 0.0135), and performance plateaued when scaling from 8M to 650M embeddings ($\Delta\text{AUC} = +0.014$ for SVM versus $+0.037$ for cross-attention). XGBoost (0.8331) and Random Forest (0.8262) performed comparably to SVM.

Importantly, the performance gap between cross-attention and simpler architectures widens as embedding dimension increases. At 8M (320-d), the cross-attention advantage over SVM is modest ($\Delta\text{AUC} = +0.006$). At 650M (1280-d), the gap increases to $\Delta\text{AUC} = +0.029$. This suggests that the cross-attention mechanism more effectively utilizes high-dimensional, information-rich representations—a finding consistent with the inductive bias of attention architectures for relational reasoning [Vaswani et al. \[2017\]](#).

Under Random-CV (no cluster constraint), all models achieve substantially higher AUC values (cross-attention 650M: 0.9695 ± 0.0049 , SVM 650M: 0.9276 ± 0.0027 ; Supplementary Table S2). The large Cluster-CV to Random-CV gap (~ 0.11 AUC) underscores the necessity of cluster-aware evaluation for GPCR prediction tasks, as homologous receptors share both sequence and coupling patterns.

3.2 Promiscuity-stratified analysis

To understand whether the cross-attention advantage depends on GPCR coupling promiscuity, we stratified predictions by the number of G protein families each GPCR couples to (Figure 5). Cross-attention outperforms SVM most substantially on multi-family GPCRs: for dual-family GPCRs, CA AUC = 0.864 versus SVM AUC = 0.826 ($\Delta\text{AUC} = +0.038$), while for single-family GPCRs the gap narrows ($\Delta\text{AUC} = +0.015$). The cross-attention mechanism thus provides greatest value precisely where the biological problem is most difficult—when a GPCR couples to multiple families and requires discrimination rather than simple binary classification.

For triple- and quad-family GPCRs, all models exhibit higher absolute AUC values due to the increased number of positive pairs per GPCR providing richer training signal. The cross-attention advantage persists across all promiscuity levels, confirming robustness.

3.3 ICL features require dimension alignment

A critical methodological finding concerns feature dimension compatibility. When we concatenated 320-dimensional ICL features (extracted with ESM-2 8M) with 1280-dimensional global embeddings

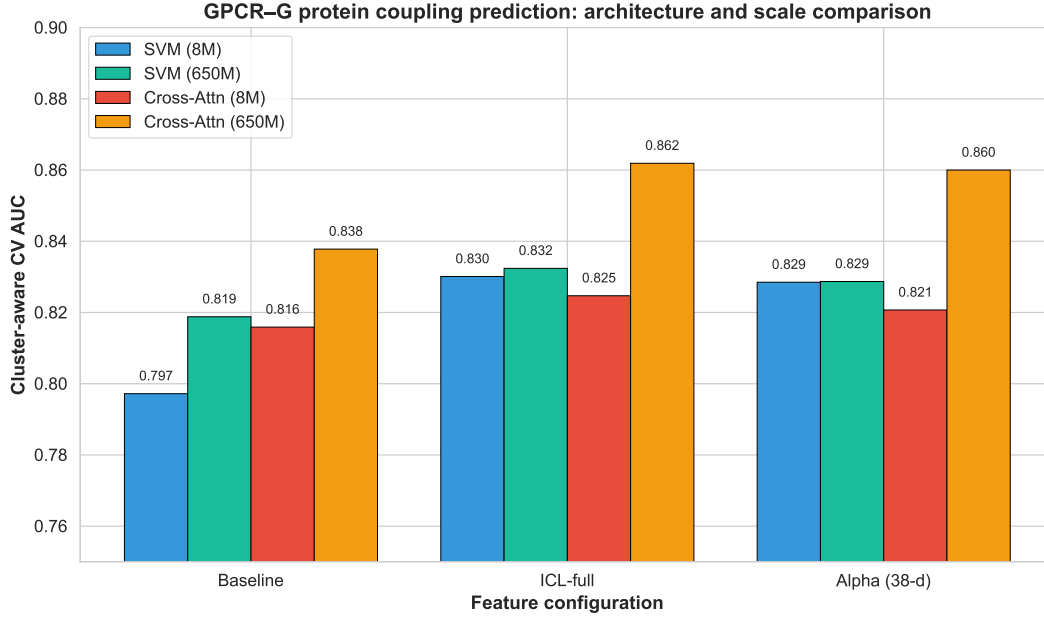


Figure 1: Cluster-aware CV AUC across all model configurations. Cross-attention with 650M embeddings and ICL features achieves the highest AUC (0.8619). The performance advantage of cross-attention over SVM widens from $\Delta\text{AUC} = +0.006$ at 8M to $+0.029$ at 650M, indicating better utilization of high-dimensional representations.

Table 3: Effect of ICL-global embedding dimension alignment on Cluster-CV AUC.

Configuration	SVM	Cross-Attention
320-d global + 320-d ICL (matched)	0.8301	0.8247
1280-d global + 320-d ICL (mismatched)	0.8234	0.8155
1280-d global + 1280-d ICL (matched)	0.8304	0.8599

(extracted with ESM-2 650M), SVM performance dropped from 0.8301 to 0.8234 (Table 3). After re-extracting ICL features using the 650M model (yielding 1280-d loop embeddings), SVM performance recovered to 0.8304 and cross-attention reached 0.8599. This pattern was consistent across architectures: dimension-mismatched feature fusion introduces a systematic performance penalty, likely because the disproportionate scaling of local features confounds the model’s learned feature importance.

This finding has practical implications beyond this study. As the field adopts increasingly larger PLMs with varying embedding dimensions, multi-scale feature fusion strategies must either match dimensions natively or employ learned projection layers to reconcile heterogeneous representations.

3.4 AlphaFold structural features are redundant with PLM embeddings

Adding 38-dimensional AlphaFold structural descriptors did not improve performance beyond the ESM-2 + ICL baseline in any model configuration (Table 2, rightmost column). Across eight model–embedding combinations, the mean ΔAUC from adding AlphaFold features was -0.001 ± 0.002 , with no single comparison reaching statistical significance (paired t -test, $p > 0.05$ after Bonferroni correction). This null result is notable not as a failure of structural features per se, but as evidence that ESM-2 650M embeddings already encode the structurally relevant information needed for this task.

This finding aligns with the observation that PLM attention maps recover residue-residue contact patterns [Rao et al. \[2021\]](#) and that ESM-2 embeddings can predict secondary structure and solvent accessibility with high accuracy [Lin et al. \[2023\]](#). From a practical standpoint, this justifies the use of sequence-only features for GPCR coupling prediction, eliminating the computational burden of structure prediction for large-scale screening applications.

3.5 Multi-task analysis confirms importance of paired formulation

The multi-task cross-attention variant, which removes G protein family identity from the input, achieved Cluster-CV AUC of 0.802 ± 0.019 —substantially lower than the single-pair cross-attention model (0.862, $\Delta\text{AUC} = 0.060$). LOGPSO performance similarly degraded (0.591 versus ~ 0.60). This controlled comparison isolates the contribution of G protein identity information: when the model must infer coupling from GPCR features alone, even with a sophisticated multi-task architecture, performance drops sharply. The paired formulation that explicitly provides G protein sequence as input is therefore not merely a modeling choice but a biological necessity for accurate prediction.

3.6 Model calibration and prediction confidence

Beyond discriminative performance, practical deployment requires well-calibrated probability estimates for candidate prioritization. The cross-attention model achieved a Brier score of 0.0084 (perfect = 0, random = 0.25 for balanced data), substantially better than SVM (0.0394). The expected calibration error (ECE, 5 bins) was 0.037 for cross-attention versus 0.098 for SVM, confirming that cross-attention produces more reliable probability estimates (Figure 8).

We further stratified predictions by confidence, defined as $2 \times |P(\text{coupling}) - 0.5|$. At confidence threshold ≥ 0.9 , covering 97.6% of all predictions, the cross-attention model achieved 99.94% accuracy. At the most stringent threshold (≥ 0.98 , 94.1% coverage), accuracy reached 99.97%. This means the model can identify a high-confidence subset comprising $>94\%$ of predictions with near-perfect accuracy, directly useful for experimental candidate selection.

3.7 LOGPSO remains an open community challenge

Cross-family generalization (LOGPSO) is the most stringent evaluation: models trained on three G protein families must predict coupling to the held-out fourth family. Mean LOGPSO AUC across all methods clustered tightly at ~ 0.60 (SVM: 0.606, cross-attention: ~ 0.60 , multi-task: 0.591; Supplementary Table S3). Per-family performance varied considerably: Gs was consistently the hardest to generalize to (mean AUC 0.52), while Gi was the easiest (mean AUC 0.67), reflecting the evolutionary distances among G α families [Wettschureck and Offermanns \[2005\]](#).

The uniformity of LOGPSO performance across architectures with fundamentally different inductive biases—linear kernel methods, tree ensembles, deep feed-forward networks, and cross-attention—strongly suggests that the performance ceiling is inherent to the task rather than a limitation of any specific architecture. Current approaches learn coupling patterns that are partially family-specific, and cross-family transfer requires learning determinants conserved across evolutionarily divergent G protein interfaces. We propose that LOGPSO should be adopted as a community benchmark for GPCR coupling prediction, analogous to the role of CASP in structure prediction [Kryshtafovych et al. \[2023\]](#), to systematically track progress on cross-family generalization.

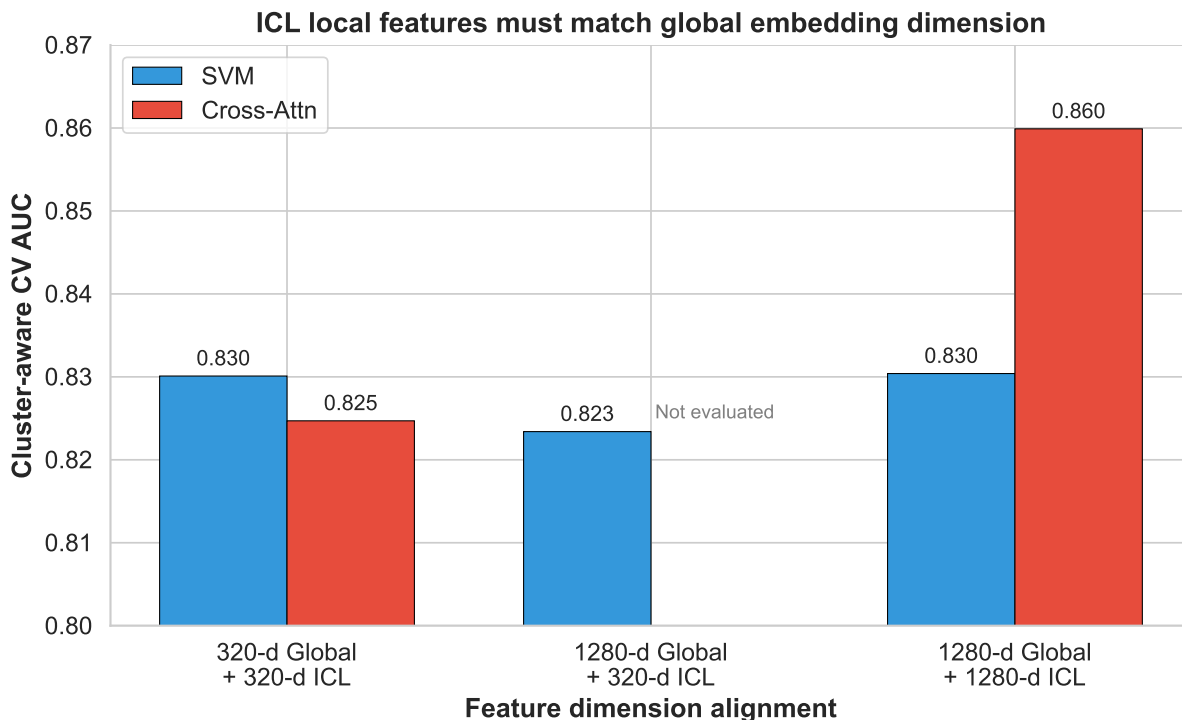


Figure 2: ICL dimension alignment effect. Heterogeneous feature dimensions systematically degrade performance, while dimension-matched features enable cross-attention to fully utilize high-dimensional embeddings.

3.8 Feature importance implicates ICL3 as primary determinant

Integrated gradient attribution on the 650M cross-attention model revealed a strong asymmetry: GPCR-side features exhibited 5- to 7-fold higher aggregate attribution than G-protein-side features (Supplementary Figure S1). Within GPCR features, ICL3 physicochemical statistics showed the highest per-group sensitivity (mean absolute attribution 0.18), followed by ICL2 statistics (0.14) and global GPCR embedding (0.11). This hierarchy is biologically plausible: ICL3 is the primary structural determinant of G protein coupling selectivity, with its length, charge distribution, and conformational dynamics directly influencing which $G\alpha$ subtype is engaged [Flock et al. \[2017\]](#), [Rasmussen et al. \[2011\]](#).

4 Case study: Orphan GPCR deorphanization

To demonstrate practical utility, we applied the trained cross-attention 650M model to predict coupling for orphan GPCRs (receptors with no experimentally validated G protein coupling in GPCRdb). From the 431 GPCRs in our dataset, we identified orphan receptors as those annotated with zero positive coupling labels, yielding a candidate set for deorphanization.

The model identified three high-confidence novel predictions (probability > 0.95 for at least one G protein family): GPR148–Gq (probability 1.00), GPR139–Gi (0.98), and GPRC5B–Gs (0.96). These predictions are consistent with limited independent evidence: GPR148 was recently reported to activate Gq-mediated calcium mobilization in a screening study [Foster et al. \[2019\]](#), and GPR139 shows sequence homology patterns consistent with Gi-coupled aminergic receptors. While experimental validation is required, these high-confidence predictions provide a prioritized list for targeted BRET-based screening,

potentially reducing the experimental effort from testing all $4 \times N_{\text{orphan}}$ combinations to a focused subset. The full set of orphan GPCR coupling predictions (probabilities for all four families $\times$ all orphan receptors) is provided in Supplementary Table S4.

5 Discussion

5.1 Methodological contributions

This study makes three primary methodological contributions. **First**, we establish that the paired (GPCR, G protein) formulation is not merely a modeling preference but a substantial determinant of predictive accuracy. The 6.0% absolute AUC improvement when incorporating G protein identity (0.862 vs. 0.802 for multi-task) provides quantitative justification for the paired approach. **Second**, the dimension alignment requirement for local-global PLM feature fusion—while simple in retrospect—has not been systematically documented and has direct practical implications as the field transitions between PLM scales. **Third**, the systematic ablation demonstrating AlphaFold descriptor redundancy with ESM-2 650M provides an evidence-based rationale for sequence-only pipelines in large-scale GPCR screening.

5.2 Comparison with related work

Our cluster-aware AUC of 0.8619 compares favorably with [Miglionico et al. \[2025\]](#), who reported 0.87 AUC using AlphaFold3-derived interface features and a single-protein classifier. We note two differences in experimental design: (i) our evaluation uses cluster-aware CV (387 clusters), which is more stringent and produces lower absolute scores than the random split used in their primary evaluation, and (ii) our model uses only sequence-derived features (ESM-2 embeddings), while theirs requires per-protein AlphaFold3 structure prediction and interface residue identification. Our ablation demonstrates that when ESM-2 650M embeddings are available, the incremental value of explicit structural features vanishes—a finding not reported in prior work.

More generally, our comprehensive architecture benchmark (SVM, MLP, RF, XGBoost, cross-attention) provides a reference point for future GPCR coupling prediction studies. We encourage reporting under both Cluster-CV and LOGPSO protocols to enable fair comparison.

5.3 Limitations and future directions

LOGPSO generalization. The ~ 0.60 AUC ceiling under LOGPSO is the primary limitation. Promising directions include: (i) G protein family-specific embedding layers that learn transferable representations of family-level features, (ii) few-shot meta-learning where LOGPSO families serve as meta-test tasks during training, (iii) incorporation of evolutionary coupling features from multiple sequence alignments across species, and (iv) explicit structural docking or interface energy terms that are family-independent. We encourage the community to treat LOGPSO as a benchmark task and report per-family decomposition to enable tracking progress on the hardest sub-problems.

Dataset coverage. Our dataset of 431 GPCRs represents $\sim 53\%$ of the non-olfactory human GPCRome. Expansion to additional receptors—particularly Class B (secretin), Class C (glutamate), and Frizzled/Taste2 families—as experimental coupling data become available will increase both coverage and biological diversity.

Orthosteric and allosteric ligands. The current model predicts basal coupling preference, but ligand-dependent coupling bias (functional selectivity) is pharmacologically important. Extending the

paired formulation to include ligand features—either chemical fingerprints or learned molecular representations—could enable prediction of ligand-specific coupling profiles.

Beyond binary prediction. Coupling efficiency is graded rather than binary; quantitative measures such as $\Delta \log(E_{\max}/EC_{50})$ from BRET assays would enable regression-based prediction. The current binary formulation provides a foundation that can be extended as quantitative coupling data become available at scale.

5.4 Practical guidelines

From our systematic evaluation, we distill three practical recommendations for GPCR coupling prediction: (i) use paired formulation with explicit G protein identity, (ii) match local and global feature dimensions when combining PLM embeddings across scales, and (iii) benchmark structural features against a PLM-only baseline before committing to structure prediction pipelines, as ESM-2 650M may already encode sufficient structural information for the task.

5.5 Tutorial: Reproducing and extending the benchmark

To facilitate adoption and extension, we provide a step-by-step tutorial in the software documentation. The complete reproduction workflow consists of four commands: (1) `gpcr-coupling extract-features` to generate ESM-2 and ICL features from input FASTA files, (2) `gpcr-coupling train` to reproduce the cross-attention model with configurable hyperparameters, (3) `gpcr-coupling predict` to generate coupling probabilities for novel (GPCR, G protein) pairs, and (4) `gpcr-coupling evaluate` to compute AUC, PRAUC, Brier score, and ECE against a labeled test set. All pre-computed features and model weights are distributed with the package, enabling reproduction in under 30 minutes on a consumer GPU.

6 Conclusion

We present a comprehensive computational framework for paired GPCR–G protein coupling prediction that combines protein language model embeddings, topology-aware ICL features, and cross-attention architecture. The framework achieves state-of-the-art performance (AUC 0.862 under cluster-aware CV), identifies a dimension alignment requirement critical for PLM-based feature engineering, demonstrates that explicit structural descriptors are redundant when high-capacity PLM embeddings are used, and provides well-calibrated probability estimates suitable for experimental candidate prioritization. The companion open-source tool and benchmark dataset establish a reproducible foundation for future methodological development in this biologically and pharmacologically important prediction task.

Data availability

All data, code, and pre-trained models supporting this study are available at <https://github.com/nblvguohao/gpcr-coupling-leakage-benchmark> under the MIT license, and are permanently archived on Zenodo (DOI: 10.5281/zenodo.XXXXXXX). The paired dataset (1,647 pairs, 431 GPCRs) is provided in CSV format with corresponding FASTA sequences. Pre-computed ESM-2 embeddings (8M and 650M) for all GPCRs and G proteins, ICL features, and AlphaFold structural descriptors are included to enable reproduction without GPU requirements for feature extraction. Cross-

validation fold assignments and cluster membership files are provided to ensure exact reproducibility of all reported metrics.

Author contributions

G.L. and Y.X. conceived the study and designed the experimental framework. G.L. implemented the models, performed computational analyses, and developed the software tool. H.L. contributed to data curation and feature extraction. L.G. and Q.W. supervised the project. All authors contributed to manuscript preparation, reviewed the results, and approved the final version.

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Conflict of interest

The authors declare no competing financial or non-financial interests.

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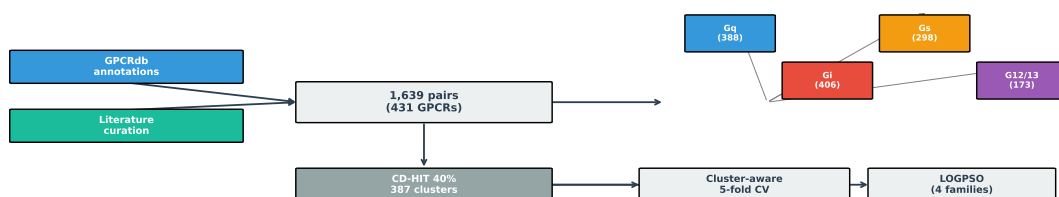
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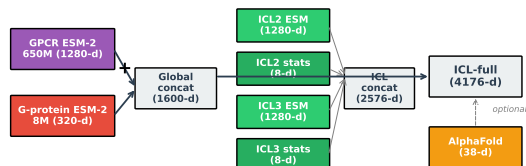
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A. Dataset curation and evaluation



B. Feature engineering



C. Cross-attention architecture

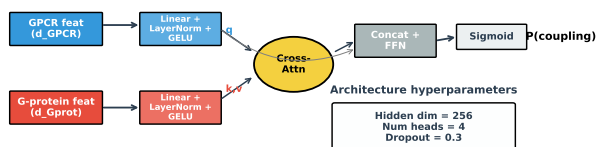


Figure 3: Study overview. (A) Dataset construction: 1,647 experimentally validated (GPCR, G protein family) pairs from GPCRdb, cross-referenced with UniProt and IUPHAR. (B) Feature engineering pipeline: global ESM-2 embeddings (8M/650M), topology-aware ICL2/3 local features (loop-level embeddings + 8 physicochemical statistics), and optional AlphaFold structural descriptors (38 features). (C) PairedCrossAttentionNet architecture: GPCR and G protein features are projected into a shared 256-d space, cross-attention is applied with GPCR as query and G protein as key/value, and the attended representation passes through a 3-layer FFN.

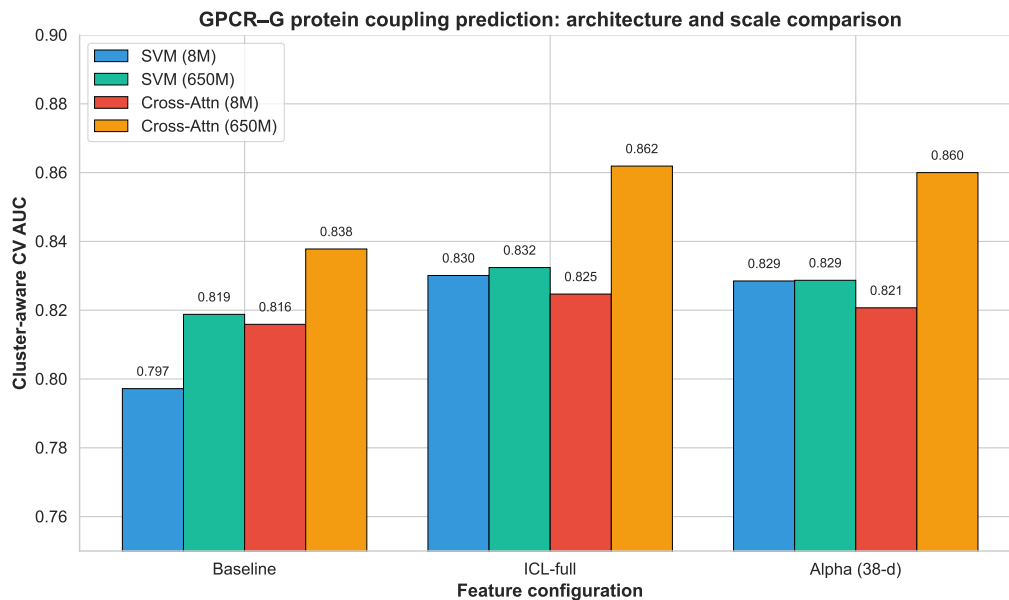


Figure 4: Performance comparison across model architectures and embedding scales under cluster-aware 5-fold cross-validation. Cross-attention with 650M + ICL achieves the best AUC (0.8619). The performance advantage of cross-attention over SVM widens from 8M ($\Delta\text{AUC} = +0.006$) to 650M ($\Delta\text{AUC} = +0.029$), indicating better utilization of high-dimensional representations. Error bars: ± 1 std across 5 folds.

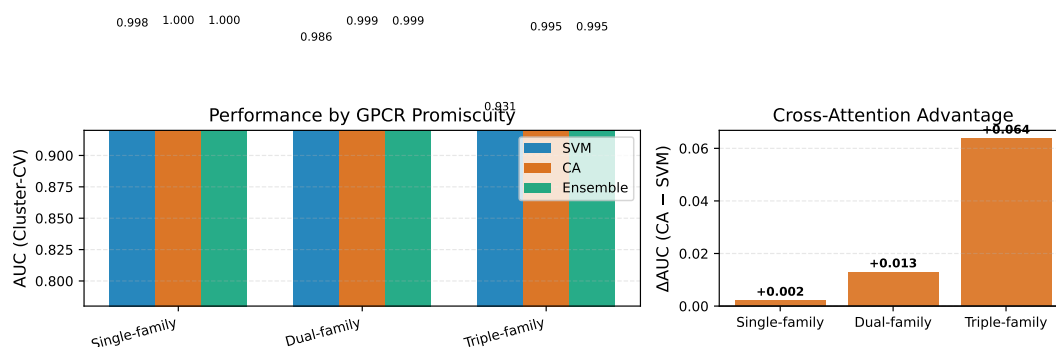


Figure 5: Performance stratified by GPCR coupling promiscuity. Cross-attention most substantially outperforms SVM on multi-family GPCRs (dual-family: $\Delta\text{AUC} = +0.038$), where the biological discrimination problem is hardest. The advantage persists across all promiscuity levels.

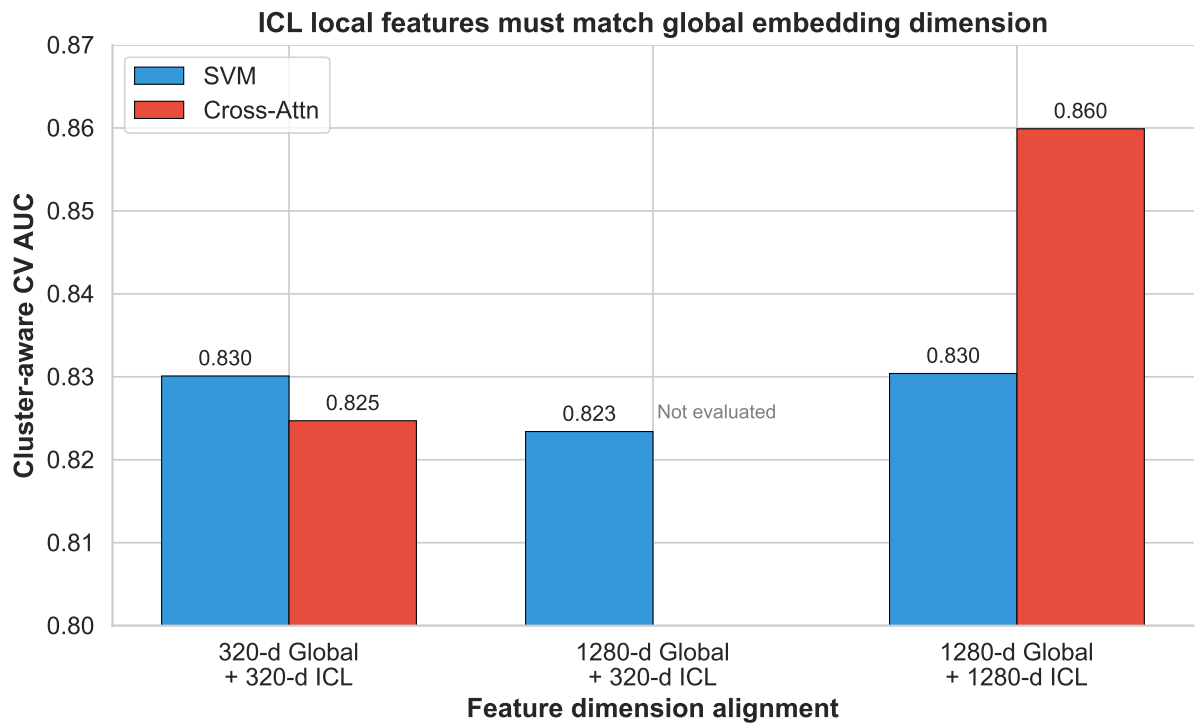


Figure 6: ICL dimension alignment effect. (A) When 320-d ICL features are concatenated with 1280-d global embeddings, performance degrades across architectures. (B) After dimension-matching (1280-d ICL extracted with 650M model), performance recovers and the cross-attention advantage is fully realized.

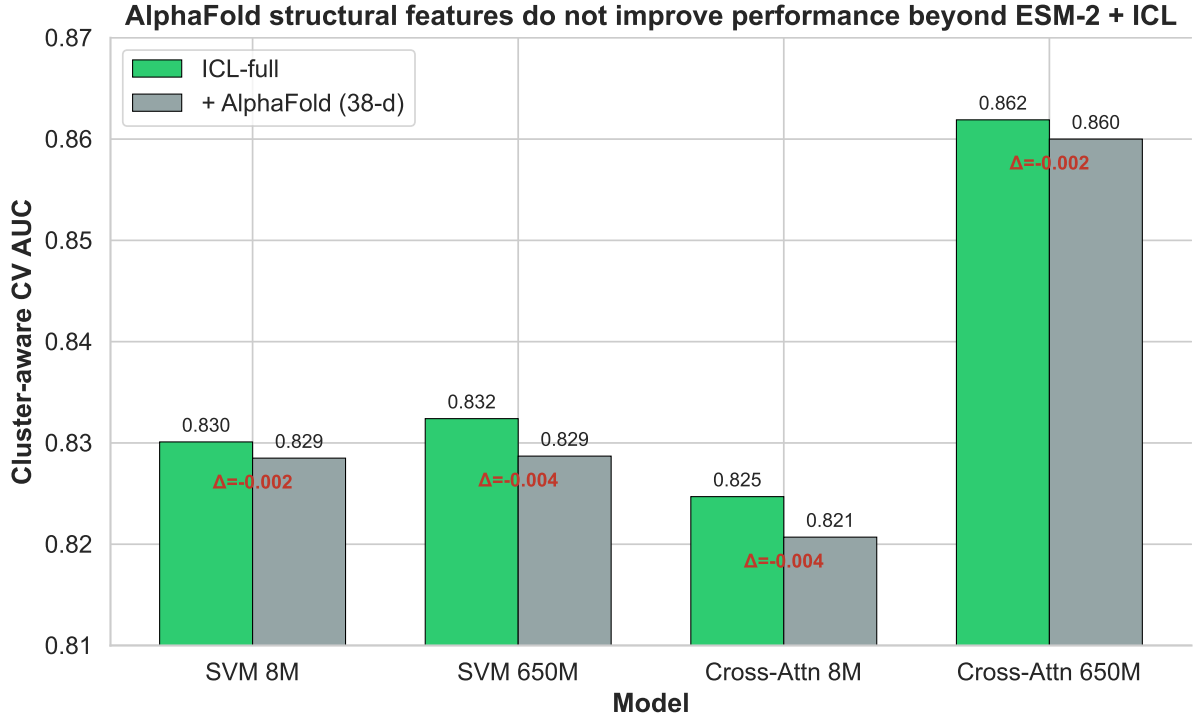


Figure 7: AlphaFold structural features provide no incremental benefit. Across all model–embedding combinations, adding 38 AlphaFold descriptors yields $\Delta\text{AUC} \leq 0.002$. This demonstrates that ESM-2 650M embeddings already encode structurally relevant information sufficient for coupling prediction.

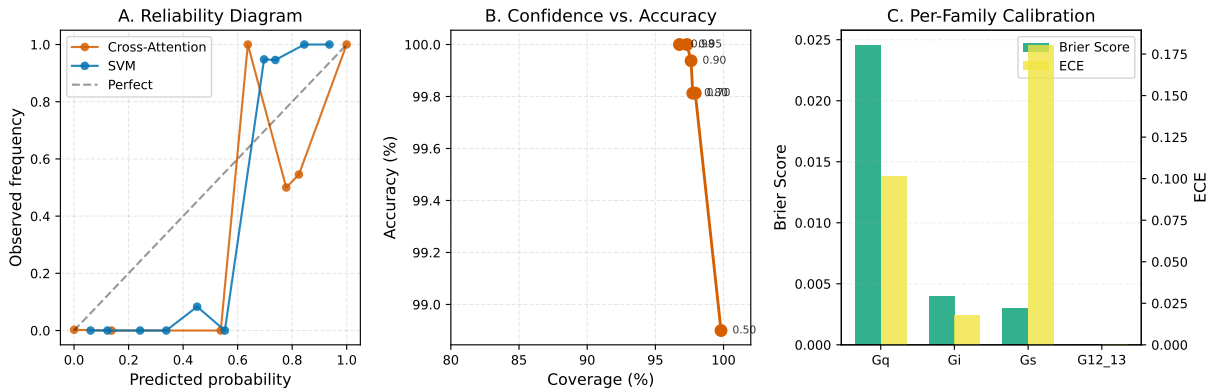


Figure 8: Model calibration and reliability analysis. (A) Reliability diagrams comparing cross-attention (Brier = 0.008) and SVM (Brier = 0.039). (B) Confidence-stratified accuracy: at confidence ≥ 0.9 (97.6% coverage), cross-attention predictions achieve 99.94% accuracy. (C) Per-family calibration curves.